

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/369841339>

Community Acquired Methicillin Resistant Staphylococci (CA-MRS) in fecal matter of Wild Birds – A ‘One Health’ Point of Concern

Article in *Journal of Infection and Public Health* · April 2023

DOI: 10.1016/j.jiph.2023.04.002

CITATIONS

10

READS

225

2 authors, including:



Abdul Rauf Tareen

Quaid-i-Azam University

8 PUBLICATIONS 13 CITATIONS

[SEE PROFILE](#)



Original article

Community acquired methicillin resistant *Staphylococci* (CA-MRS) in fecal matter of wild birds – A ‘one health’ point of concern

Abdul Rauf Tareen, Rabaab Zahra *

Department of Microbiology, Faculty of Biological Sciences, Quaid-i-Azam University, 45320 Islamabad, Pakistan

ARTICLE INFO

Article history:

Received 9 January 2023

Received in revised form 24 March 2023

Accepted 3 April 2023

Keywords:

Birds

Staphylococci

Disc diffusion

MDR

Typing

Biofilm

ABSTRACT

Background: Antibiotic resistance in Staphylococci, particularly methicillin resistance is a major public health concern. While this problem has been reported from the clinical settings, its presence in non-clinical settings also needs to be investigated. The role of wildlife in carrying and disseminating the resistant strains has been established in different studies but its role in Pakistani environment has not been explored yet. To evaluate this, we investigated the carriage of antibiotic resistant Staphylococci in wild birds from Islamabad region.

Methodology: Birds fecal matter were collected during September 2016–August 2017 from eight different environmental settings of Islamabad. Prevalence of Staphylococci, their susceptibility profile against eight classes of antibiotics through disc diffusion method, their SCCmec types, co-resistance of macrolide and cefoxitin through PCR assay and biofilm formation through microtitre plate assay were studied.

Results: Out of 320 birds feces collected, 394 Staphylococci were isolated, where 165 (42%) were resistant to at least one or two classes of antibiotics. High resistance was found against erythromycin (40%) and tetracycline (21%) while cefoxitin resistance was 18% and vancomycin resistance was only in 2%. One hundred and three (26%) isolates exhibited multi-drug resistance (MDR) pattern. *mecA* gene was detected in 45/70 (64%) cefoxitin resistant isolates. Community acquired methicillin resistant Staphylococci (CA-MRS) were 87% while Hospital acquired methicillin resistant Staphylococci (HA-MRS) were 40%. In the MRS isolates showing co-resistance to macrolides, *mefA* (69%) and *ermC* (50%) genes were more prevalent. Strong biofilm formation was observed in 90% of the MRS, of which 48% were methicillin resistant *Staphylococcus aureus* (MRSA) isolates while 52% were methicillin resistant coagulase negative Staphylococci (MRCoNS).

Conclusion: Occurrence of methicillin resistant strains of Staphylococci in wild birds suggests their role in the carriage and dissemination of resistant strains into the environment. The findings of the study strongly recommend the monitoring of resistant bacteria in wild birds and wildlife.

© 2023 The Authors. Published by Elsevier Ltd on behalf of King Saud Bin Abdulaziz University for Health Sciences. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Introduction

Staphylococci being opportunistic pathogens of humans and animals' normal commensal flora colonize about 30–50% of human population and are regarded as one of the clinically important bacterial genera. The genus colonizes mucous membrane, skin, and skin glands and thus produces a wide range of humans and animals associated infections such as rashes, meningitis, bone inflammation, septicemia, osteomyelitis, toxic shock syndrome, infective endocarditis, respiratory and skin infections [1,2]. Staphylococci are placed into two groups: coagulase positive *Staphylococcus aureus* and

coagulase negative Staphylococci (CoNS) that include important medical pathogens like *Staphylococcus epidermidis*, *Staphylococcus saprophyticus*, *Staphylococcus haemolyticus*, *Staphylococcus hominis* and 49 other species. Both groups of *Staphylococcus* genus have been linked with increased mortality rate in clinical settings [3,4,35].

Survival and adaptation abilities of Staphylococci into clinical and environmental settings and the variety of virulence factors it encodes have made Staphylococci an exceptionally successful pathogen [5]. Penicillin and its derivatives especially methicillin have been applied in the treatment of Staphylococcal infections both in humans and in animals [6]. But, with the increasing applications of these antibiotics in the clinics, Staphylococci species have developed resistance and have emerged as a serious threat to the available clinical antibiotic regimen used in the treatment of these infections [7].

* Corresponding author.

E-mail address: rzahra@qau.edu.pk (R. Zahra).

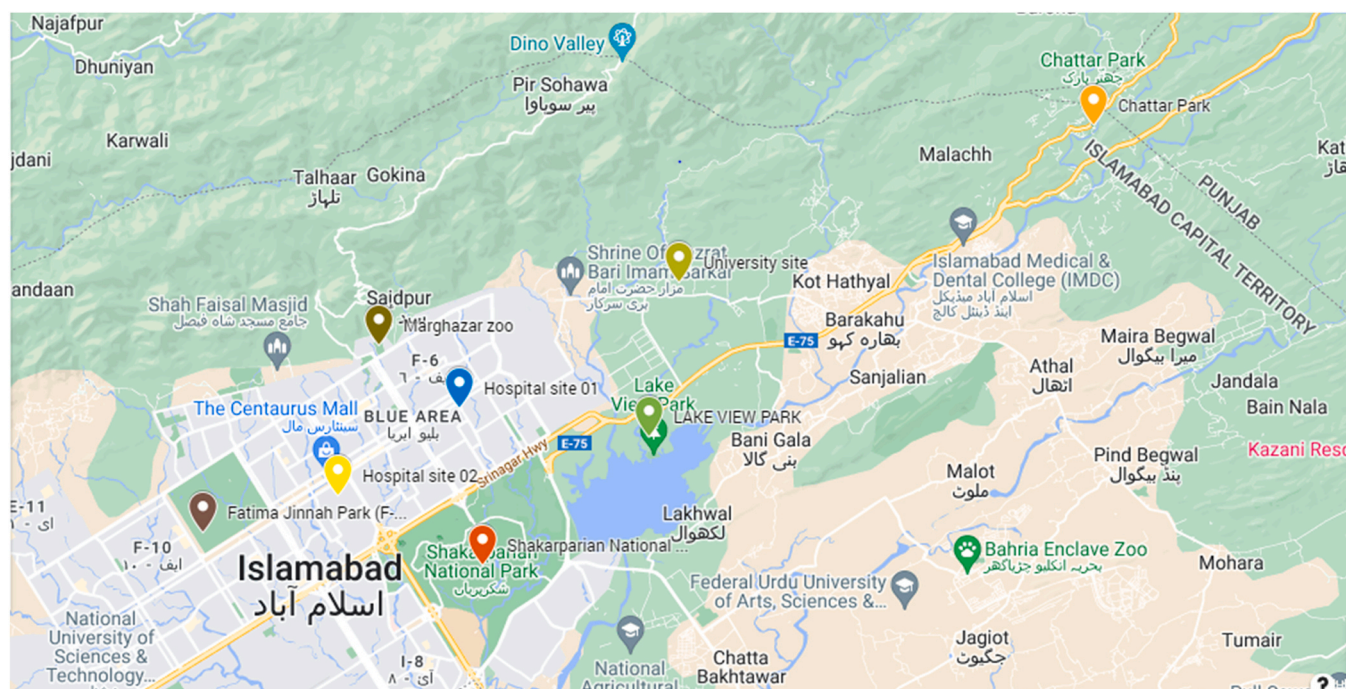


Fig. 1. Part of Islamabad city map indicating study sampling sites.

Methicillin resistance was first reported in *Staphylococcus aureus* in 1961 and is due to *mecA* gene which encodes the penicillin binding proteins (PBPs) implicated in inactivating the *beta*-lactam group of antibiotics [9]. Penicillin binding protein has enabled *Staphylococcus aureus* to evolve robust resistance mechanisms against clinically vital antibiotic classes i.e. penicillins, glycopeptides, macrolides, quinolones that are the treatment of choice in Staphylococcal infections. The frequency and versatility of Staphylococcal survival in various environments has changed the epidemiology of MRSA over the past 30 years. Once mainly regarded as hospital acquired MRSA (HA-MRSA), efflux of methicillin resistant bacteria and associated resistance genes outside the nosocomial settings into the environment facilitated the emergence of community associated MRSA (CA-MRSA). CA-MRSA are well studied for their robust transmission in the community and ability to develop aggressive and severe skin infections of the soft tissues and community-acquired pneumonia [8,9].

One health concept is a modern approach towards bringing scientific communities to safeguard humans, animals, and the environmental health. Antibiotic resistance is linked to all these components due to the irrational use in various sectors that include human medicine, aquaculture, fish farming, animal husbandry and agriculture. Due to the presence of excessive selective pressure in the environment, the bacterial communities acquire resistance determinants and increased abilities to disseminate them to humans, animals, and natural environment [10].

Wildlife is a significant reservoir of bacterial communities that are constantly exposed to resistant bacteria or resistance genes in the environmental settings through human associated activities [11]. This transmission of resistant bacteria and their respective genes among wildlife is mainly due to its interaction with the human settings that include human wastes, sewage water treatment plants (STPs), water bodies and farms which serve as a prime habitat for small animals, mammals and birds [12,13]. The extensive use of antibiotics in clinical and veterinary settings for the treatment of infectious bacterial diseases and in the animal husbandry have resulted in the emergence of diverse resistance mechanisms in

environmental bacterial communities thus compromising the antibiotics efficacy [14,15].

This study aimed to investigate the role of wild birds in the carriage and dissemination of antibiotic resistant strains of Staphylococci in various environmental settings of Islamabad, Pakistan. For this, we evaluated the antibiotic susceptibility profiles, prevalence of *mecA* gene, *SCCmec* types, macrolide resistance genes and biofilm formation ability in Staphylococci isolated from the wild birds' fecal droppings.

Materials and methods

Study design

This was a descriptive, cross-sectional study and a detailed flow chart of work has been shown in the [supplementary Figure 1](#).

Study inclusion and exclusion criteria

Inclusion: We aimed to study the prevalence and levels of resistance among the Staphylococci, therefore we included the Staphylococcal isolates and studied their details in the study.

Exclusion: Non-Staphylococcal isolates were excluded.

Samples collection

Birds' fecal droppings from eight pre-defined birds nesting sites including recreational parks (Lake view park, Shakarparian park, Chathar park, Fatima Jinnah Park, Marghazar park), university site, hospital waste dumps sites1 and 2 (Fig. 1) were collected from September 2016 to August 2017 during four seasons: autumn (September–October 2016), winter (December 2016–January 2017), spring (March–April 2017) and summer (July–August 2018). A total of 320 fecal samples (40 from each sampling site) were collected ([Supplementary Table 1](#)).

Fresh fecal matters of the birds were collected using a sterile spatula and added into 1 ml of isotonic normal saline solution in

sterile Eppendorf tubes. Maximum care was taken to avoid collecting soil sediments. Spatula was disinfected with 70% ethanol repeatedly to avoid cross contamination of fecal bacteria. Eppendorf tubes containing fecal droppings were placed in sampling box containing an ice pack and were then transported to the laboratory within one hour.

Bacterial isolation & Identification

Fecal droppings were serially diluted in normal saline solution and were then swabbed on the Mannitol salt agar (MSA) plates. MSA plates were then incubated overnight at 37 °C. All Staphylococci colonies obtained on MSA plates were isolated, purified and preserved in 40% glycerol. All the isolates were identified as either *Staphylococcus aureus* or coagulase negative Staphylococci (CoNS) using biochemical tests including tube coagulase, DNase, catalase tests and PCR based detection of *nucA* gene.

Phenotypic antibiotic susceptibility profile

Staphylococcal isolates were tested against a panel of eight antibiotics commonly prescribed in animals and humans bacterial infections treatment in Pakistan using Kirby Bauer disc diffusion method. The antibiotics were cefoxitin (30 µg), erythromycin (15 µg), gentamicin (10 µg), ciprofloxacin (30 µg), chloramphenicol (30 µg), linezolid (30 µg), nitrofurantoin (15 µg), tetracycline (30 µg), and vancomycin (30 µg). Overnight bacterial colonies were suspended in nutrient broth and their turbidity was adjusted according to the 0.5 McFarland solution. The prepared inoculum was swabbed uniformly on the surface of Mueller Hinton agar (MHA) plates and antibiotic discs were applied using antibiotic disc dispenser within 15 min [44]. The plates were incubated for 24 h at 37 °C. Zones of inhibition were measured and interpreted according to the Clinical Standard Laboratory Institute (CLSI) guidelines 2017. *Staphylococcus aureus* ATCC 25923 was used as a quality control strain.

Phenotypic confirmation of Methicillin resistant Staphylococci

Staphylococcal isolates exhibiting the zone of inhibition ≤ 21 mm to cefoxitin according to the CLSI guidelines were considered as methicillin resistant Staphylococci (Supplementary Figure 2).

PCR based detection of *nucA* & *mecA* genes

Genomic DNA was extracted by Gills DNA extraction protocol [33]. *nucA* gene was screened in Staphylococcal isolates to differentiate between *Staphylococcus aureus* and CoNS. Phenotypic MRS isolates were screened for the presence of *mecA* gene. Primer sequences were adopted from already published literature (Table 1). PCR assays were conducted in a 25 µl reaction mixture containing 12.5 µl 2 X Master mixes (Dreamtaq green Master Mix, Thermo Scientific), 1 µl each forward and reverse primers listed in Table 1. PCR cycling conditions consisted of initial denaturation at 95 °C for 10 min, final denaturation at 95 °C for 40 s, annealing temperature at 56 °C for *nucA* and 52 °C for *mecA*, initial extension at 72 °C for 1 min for 38 cycles, followed by a final extension at 72 °C for 10 mins. PCR products were loaded on 1.5% agarose gel and run for 60 min at 110volts in gel tank containing TBE buffer and visualized in gel documentation system.

SCCmec typing of Methicillin resistant isolates

Phenotypically positive MRS were further screened to identify Hospital acquired methicillin resistant Staphylococci (HA-MRS) and community acquired methicillin resistant Staphylococci (CA-MRS). For this purpose, a set of six staphylococcal cassette chromosome

(SCC) types I, II, III, IV, V, VI and Penton valentine leukocidine (PVL) primers were used (Table 1). Two triplex assays; one consisting of types I-III primers (For HA-MRS) and other of types IV-VI (CA-MRS) were performed. The PCR conditions for 25 µl reaction were: Initial denaturation at 94 °C for 5 min, final denaturation at 94 °C for 30 s, annealing temperature at 52 °C and 53 °C for 40 s, initial extension at 72 °C for 1 min and final extension at 72 °C for 10 min with 35 cyclic repetitions. Singleplex assay were run for the screening of PVL gene for which the PCR conditions were: 94 °C for 5 min, 94 °C for 45 s, 51 °C for 40 s, 72 °C for 1 min and final extension at 72 °C for 10 min with 35 cyclic repetitions.

Macrolide resistance genes screening

MRS isolates that had exhibited phenotypic co-resistance to erythromycin (representative of macrolide class of antibiotics) on susceptibility testing were studied for the presence of macrolide resistance genes *ermA*, *ermB*, *ermC*, *msrA* and *mef* genes through singleplex PCR assays (Table 1). PCR conditions adopted were; 95 °C for 5 min, 95 °C for 45 s, annealing temperature of 54 °C for *ermA*, *ermC*, *msrA*, *mef* and 57 °C for *ermB*, 72 °C for 2 min and final extension at 72 °C for 10 min with 35 cyclic repetitions.

Biofilm assessment

All 103 MDR-Staphylococci that also contained 70 MRS isolates were assessed for biofilm forming abilities using an already established protocol [22] through Microtitre plate assay. The tested isolates' optical density (OD) was measured through Multiskan spectrophotometer at 590 nm. OD was measured using the following formula:

ODc: Average OD of negative control + (3 x SD of negative control).

ODisolate: Average OD of isolates – ODc.

ODc stands for the optical density of the control used during the study. Higher OD of the well containing test isolates than that of the test isolates were considered to be biofilm producers. Following interpretation criteria was followed:

OD $\leq$ ODc=no biofilm producer (0).

OD < ODc $\leq$ 2X ODc=weak biofilm producer (+ or 1).

2XODc < OD $\leq$ 4XODc=moderate biofilm producer (++ or 2).

4XODc < OD=strong biofilm producer (+++ or 3) [30].

Results

Morphological and biochemical identification of Staphylococci

Based on colony morphology on MSA medium, 394 Staphylococci were isolated. These were further tested using biochemical tests, in which 174 (44%) isolates were positive for all three tests i.e. catalase, tube coagulase and DNase along with mannitol fermentation on MSA agar plates and were categorized as *Staphylococcus aureus*. While 220 (56%) isolates tested positive for catalase but negative for tube coagulase and DNase tests with variable results for mannitol fermentation were grouped as coagulase negative Staphylococci (Supplementary Figure 3).

nucA gene based identification of Staphylococcus aureus

All 174 isolates biochemically identified as *Staphylococcus aureus* was confirmed genotypically by the detection of *nucA* specific gene (Supplementary Figure 4).

Antibiotic susceptibility profiles

A total of 394 Staphylococci strains were isolated from 320 fecal samples, of which 74 (19%) were collected from Hospital site 1, 69

Table 1
Nucleotides sequences of the study primers.

S. No.	Primer Name	Nucleotide Sequence (5'–3')	Amplicon Size	References
1.	<i>mecA</i>	F: TGCTATCCACCCTCAAACAGG R: AACGTTGTAACCAACCAAGA	285 bp	[16]
2.	<i>nucA</i>	F: GCGATTGATGGTGATACGGIT R: AGCCAAGCCTTGACGAACATA AGC	295 bp	[35]
3.	<i>PVL</i>	F: CAGGAGGTAATGGTTCAITTT R: ATGTCCAGACATTTTACCTAA	151 bp	[17]
4.	SCCmec Type I	F: TTCGAGTTGCTGATGAAGAAGG R: ATTTACCACAAGGACTACCAGC	495 bp	[18]
5.	SCCmec Type II	F: AATCATCTGCCATTGGTGATGC R: CGAATGAAGTGAAGAAAGTGG	284 bp	[18]
6.	SCCmec Type III	F: TTCTTAAGTACACGCTGAATCG R: GTCACAGTAATTCATCAATGC	414 bp	[19]
7.	SCCmec Type IV	F: AGTTTCTCAGAATTGCAACG R: AGTTTCTCAGAATTGCAACG	311 bp	[19]
8.	SCCmec Type V	F: TTCTCCATTCTGTTCATCC R: AGAGACTACTGACTTAAGTGG	377 bp	[18]
9.	SCCmec Type VI	F: CATCCTATGATAGCTTGGTC R: CTAATCATAGCCATGACCG	342 bp	[18]
10.	<i>ermA</i>	F: TATCTTATCGTTGAGAAGGGATT R: CTACACTTGGCTTAGGATGAAA	139 bp	[20]
11.	<i>ermB</i>	F: CTATCTGATTGTTGAAGAAGGATT R: TTTACTCTTGGTTTAGGATGAAA	142 bp	[20]
12.	<i>ermC</i>	F: CTTGTTGATCAGGATAATTTCC R: ATCTTTTAGCAAACCCGTATTC	190 bp	[20]
13.	<i>msrA</i>	F: TCCAATCATTCACAAAAATC R: AATTCCCTCTATTGGTGGT	163 bp	[21]
14.	<i>mefA</i>	F: AGTATCATTAATCACTAGTGC R: TTCTCTGGTACAAAAGTGG	348 bp	[21]

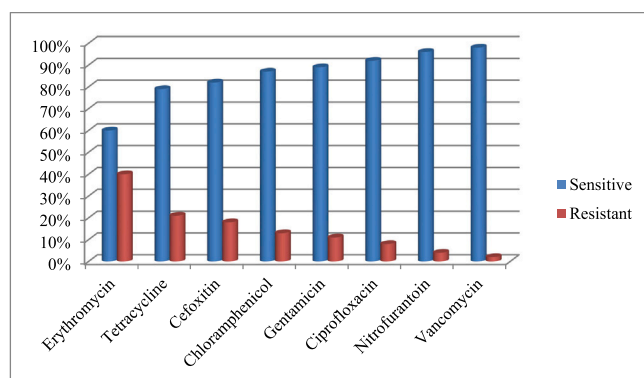


Fig. 2. Antibigram profiles of Staphylococci.

(18%) from Hospital site 2, 65 (16%) from Lake view park, 51 (13%) from chathar Park, 38 (10%) from university site, 36 (9%) Fatima Jinnah park and Marghazar zoo each, and 25 (6%) were collected from Shakarparian park. One hundred and sixty-nine (43%) isolates were sensitive to all antibiotics tested while 225 (55%) exhibited resistance to either one or more classes of the antibiotics. Resistance among Staphylococci was found to be high against erythromycin (40%) and tetracycline (21%). Resistance was also observed against cefoxitin (18%), chloramphenicol (13%), gentamicin (11%), ciprofloxacin (8%), linezolid (5%) nitrofurantoin (4%) and vancomycin (2%). Detailed data regarding the antibiotic susceptibility profile is presented in Fig. 2.

However, 103 (26%) isolates of Staphylococci were found to be multi-Drug resistant on the basis of their resistance to at least three or more classes of antibiotics [41]. Among these, 48% were *Staphylococcus aureus* and 52% were coagulase negative Staphylococci. Highest percentage of MDR isolates were isolated from the fecal samples from hospital site 1 (16%) and Fatima Jinnah Park (15%), followed by hospital site 2 (14%), University site (14%), Lake view and Shakarparian parks (13% each), Marghazar Park (10%) and Chathar park (8%) (Fig. 3).

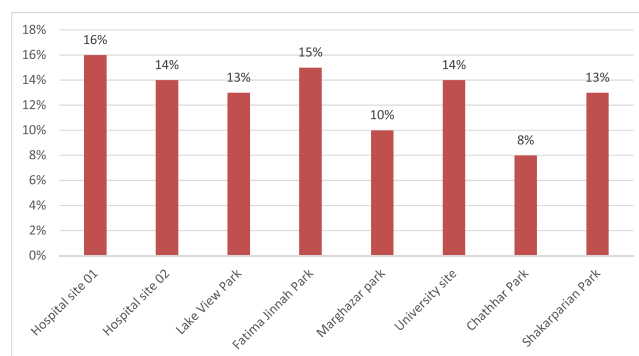


Fig. 3. Prevalence of MDR-Staphylococci isolated from different sampling sites.

Seasonal antibiotic susceptibility profile

A total of 394 Staphylococci were isolated during four seasons i.e. autumn (n=95), winter (n=127), spring (n=89) and summer (n=83). Isolates from different seasons tested against eight antibiotics classes exhibited varying degree of resistance pattern. Staphylococci from autumn and winter seasons exhibited increased resistance as compared to other seasons' isolates. Seasonal resistance during these two seasons was highest against erythromycin (34% in autumn, 57% in winter), followed by tetracycline (16% and 47%) and chloramphenicol (9% and 26%). Resistance to cefoxitin was high in autumn isolates (25%), almost similar in winter (17%) and spring isolates (18%) while comparatively low in summer isolates (10%). Vancomycin resistance was generally low in all seasons' isolates (5% in summer, 2% in spring, 1% in winter isolates). Ciprofloxacin resistance (16%) was particularly high in winter and resistance to gentamicin (16%) was found to be high in autumn Staphylococci. Lowest resistance was observed against nitrofurantoin for which resistance levels among all seasonal isolates were similar. Among all classes of antibiotics tested, linezolid was observed as the only group that had the lowest resistance pattern during autumn and winter but had highest resistance during spring (15%) and summer (10%). Detailed seasonal antibiotic resistance

profile of the Staphylococci is presented in [supplementary figure 5 and Table 4](#).

Seasonal prevalence of MDR-Staphylococci. Prevalence of MDR-Staphylococci varied among the seasonal isolates. Out of 103 MDRs isolates, 41 (40%) were isolated from winter samples, followed by Staphylococci from autumn 26 (25%). Spring (22%) and summer (13%) Staphylococcal isolates had low prevalence of MDRs isolates ([Supplementary figure 6](#)).

Phenotypic Methicillin resistance

Seventy out of 394 (18%) isolates resistant to cefoxitin were considered phenotypic MRS isolates out of which, 36 (51%) were *Staphylococcus aureus* and were designated as MRSA and 34 (49%) were coagulase negative Staphylococci, designated as MRCoNS. The resistance levels of methicillin resistant Staphylococci studied against other antibiotics classes were: erythromycin (61%), gentamicin (46%), ciprofloxacin (26%), tetracycline (47%), linezolid (11%), chloramphenicol (26%) and nitrofurantoin (1%) ([Supplementary figure 7](#)).

Genotypic analysis

mecA detection and SCCmec typing analysis

Seventy phenotypic MRS isolates were screened for the presence of *mecA* through PCR using sequence specific primers. *mecA* gene was detected in 45 (64%) of which 23/45 (51%) were *Staphylococcus aureus* while 22/45 (49%) were MRCoNS ([Supplementary figure 8](#)). SCCmec typing of the *mecA* positive isolates revealed type VI as the most prevalent SCCmec type in 19/45 (42%) followed by type V in 17/45 (38%), type IV and type II (7%, $n=3$ each), type I (4%, $n=2$) and type III (2%, $n=1$) of the MRS isolates. Based on SCCmec types, 6 (13%) of the MRS were identified as HA-MRS while 39 (87%) were identified as CA-MRS. The prevalence of HA-MRS was equal in both MRSA and MRCoNS (3/6, 50%) each. Sixty percent (23/39) of CA-MRS were MRSA while 40% (16/39) were MRCoNS ([Table 2](#)). PVL gene was observed in only 3/45 (7%) isolates, all of which were *Staphylococcus aureus*.

Co-resistance of erythromycin and cefoxitin in MRS isolates

Thirty-five (50%) out of 70 MRS isolates exhibited co-resistance to erythromycin. Highest prevalence of *mefA* was in 24/35 (69%) isolates, followed by *ermC* (50%, $n=18$), *msrA* (31%, $n=11$), *ermB* (15%, $n=5$) and *ermA* in only 4% ($n=1$) isolates. Macrolide resistance was high in MRSA isolates 19/35 (54%) while 16/35 (46%) in MRCoNS. *mefA* was the most prevalent resistance gene in both MRSA (58%) and MRCoNS (50%) followed by *ermC* (47% and 38%), *msrA* (21% and 38%) and *ermB* (21% and 13%). *ermA* was detected in only 1 isolate of *Staphylococcus aureus* ([Supplementary table 2](#)).

Statistical analysis. Z-test analysis was performed to find the statistical significance of macrolide resistance. The test results showed the Z test value for MRSA to be 6.409911 and MRCoNS was 5.460294. We carried out one-tailed test of proportion under 5% level of significance. The test result showed that the Z test value is

13.06345 which is significantly deviant from the critical region of +1.645. This shows that the macrolide resistance was significantly high in MRSA isolates than the MRCoNS.

Assessment of Biofilm formation

Biofilm formation in MDR-methicillin sensitive Staphylococci (MDR-MSS)

Out of 103 MDR isolates, 33 MDR-MSS isolates produced varying degrees of biofilm formation: 2/33 (6%) isolates showed no biofilm activity, 1(3%) of the isolates produced weak biofilm, 3 (9%) were moderate while 27(82%) isolates produced strong biofilm ([Supplementary table 3](#)). Among the strong biofilm formers, 15 (56%) were MDR-CoNS and 12 (44%) were MDR-*Staphylococcus aureus*.

Biofilm formation in MDR-MRS

Among MDR-MRS 2 (3%) showed no biofilm activity, 3 (4%) were weak, 2 (3%) moderate while 63 (90%) were strong biofilm formers. Among the strong biofilm formers, 30 (48%) were MRSA while 33 (52%) were MRCoNS isolates. The distribution of strong biofilm former isolates from different sampling sites was as: Fatima Jinnah Park 13/63 (21%), Marghazar zoo 11 (17%), University site 10 (16%), Shakarparian park and hospital site 1, 9 (14% each), Hospital site 25 (8%), Lake view park 3 (5%), and Chattar park 2 (3%) ([Supplementary figure 9](#)).

Statistical analysis. Z-test analysis was performed to measure the statistical significance of biofilm formation in MRSA and MRCoNS. The test results showed that the Z test value for MRSA was 7.625867 and MRCoNS was 8.261356. We carried out one-tailed test of proportion under 5% level of significance. The test result showed the Z test value to be 16.77546 which is significantly deviant from the critical region of +1.645. This showed that the MRCoNS biofilms formation ability was significantly high than MRSA isolates.

Discussion

Limited information is available regarding the resistance of Staphylococci in wild birds globally, and majority of the studies have focused on poultry and domestic birds [24,38,42,43]. In the current study, resistance in Staphylococcal isolates was observed against the entire panel of tested antibiotics, among which resistance was higher against erythromycin (40%), tetracycline (21%) and cefoxitin (18%). Resistance studied against chloramphenicol (13%), gentamicin (11%), ciprofloxacin (8%) and linezolid (5%), nitrofurantoin (4%) and vancomycin (2%) were relatively low. A very handful number of studies have reported resistance in wild birds and are partially similar to our study results in terms of similar Staphylococcal resistance reported such as against flouroquinolones (8% & 10%), chloramphenicol (15%) gentamicin (6%), linezolid (2% & 5.5%), but have reported higher resistance against erythromycin (40%) tetracycline (61% & 80%) and oxacillin (77%, 81% & 83.6%) [25,26,38].

Methicillin resistant Staphylococci (MRS) have emerged as a global public health concern in both human and veterinary settings. MRS being opportunistic pathogens have zoonotic roots and can cause a variety of infections that are fatal both to human and animal health. The emergence of methicillin resistance and its dissemination has narrowed down the treatment options to the last line class of glycopeptides and to synthetically manufactured antibiotics such as linezolid [40]. We detected 18% phenotypic and 64% genotypic methicillin resistance among the Staphylococci isolates. MRS phenotypic prevalence (25%) in a previous study conducted on ornamental birds [25] is similar with our results while MRS prevalence in our study is considerably lower than those reported from magpies and vultures (87%) in Spain [32], and feces of urban birds (81%) in Netherlands [26]. Few other studies have reported comparatively

Table 2
SCCmec type's distribution among different categories of Staphylococcal isolates.

SCCmec types	MRSA	MRCoNS	HA-MRS	CA-MRS
Type I	1	1	2	0
Type II	2	1	3	0
Type III	0	1	1	0
Type IV	2	1	0	3
Type V	10	7	0	17
Type VI	11	8	0	19
Total	26 (58%)	19 (42%)	6 (13%)	39 (87%)

lower MRS prevalence in wild birds of Austra-Czech republic (0.3%) [30], in white stork nestlings (3.3%) [31] and from the feces of geese from Ohio, USA (3.8%) [27]. We observed at least 2 folds increased resistance to the tested antibiotics in MRS isolates: Erythromycin (61%), gentamicin (46%), ciprofloxacin (26%), tetracycline (47%), chloramphenicol (26%), linezolid (11%).

mecA gene is situated on a mobile genetic element known as staphylococcal cassette chromosome (SCCmec) [40]. The MRS that emerges in the settings other than the hospital settings is termed as CA-MRS and is known for its rapid transmission among community and is differentiated from HA-MRS by the presence of SCCmec types VI, V, VI and Pentone valentine leukocidine (*PVL*) genes [9]. SCCmec typing analysis of the MRS isolates of the study revealed that the most prevalent SCCmec type was SCCmec type VI detected in majority of the study isolates, hence 87% of the study isolates were identified as CA-MRS. Higher number of CA-MRS were *Staphylococcus aureus* than CoNS. Three *Staphylococcus aureus* isolates were found to carry *PVL* gene. *PVL* and SCCmec types V and VI are prominent molecular markers of CA-MRS [34]. Their presence in the feces of wild birds is a major concern as it clearly indicates the dissemination of MRS into various environmental settings.

Macrolides are clinically important class of antibiotics applied in the treatment of infections caused by MRS isolates and are opted as major alternatives to β -lactams antibiotics. The evolution of macrolide resistance among MRS isolates is associated with various genotypic resistance mechanisms and have been reported [39]. In the current study, *mefA* and *ermC* macrolide resistance genes were found to be most associated with macrolide resistant MRS isolates. MRSA isolates were found to be more linked with macrolide resistance than MRCoNS. Staphylococci are known for their biofilm production and biofilm-related infections. The biofilms are organized communities of bacteria that are embedded in a complex matrix of proteins, polysaccharides and result in reducing the effectiveness of antimicrobials and persistence of various bacterial infections [34]. We found majority of the MDR isolates of the study as strong biofilm formers: 82% in MDR-MSS and 90% in MDR-MRS Staphylococci. Among MDR-MRS, biofilm production was found to be statistically more significant in MRSA than MRCoNS. Antibiotic resistance and biofilms in Staphylococci are linked functionally, as the biofilm expressed phenotype is influenced by the acquisition of antibiotic resistance. Biofilm formation in Staphylococci with increasing resistance is thus an emerging public health concern [34].

The study included eight different sampling sites which consisted of recreational parks and hospital waste dumps sites. Almost similar resistance was observed both from the environmental settings (12%) and hospitals settings (15%), indicating that the resistant strains of Staphylococci were equally prevalent in various environmental settings. It is generally understood that the spread of resistant strains originates and disseminate into the clinical settings but the occurrence of such strains in the environment highlights few loopholes which triggers this phenomenon. First, the efflux of the resistant strains from the hospital settings and secondly the unchecked and unmanaged human activities which include improper disposal of domestic, industrial waste, irregular use of antibiotics in the growth promotion of farmed and domesticated animals have played a significant role to add these 'biological resistant pollutants' in the environment. Such contaminated environment may have affected the wild birds and contributed to the transmission of resistant strains and genes in them, and as a result birds are acting as carriers and disseminators of these resistant phenotypes among various environmental settings upon their interaction for food uptake. The wild birds due to their freely mobile nature can transfer these resistant elements to bird population or to the human associated settings regardless of any geographical limitations. Wild birds' role in the carriage of resistant strains must be expanded in future

research studies for better monitoring of our environment pollution with the unwanted phenomenon of antibiotic resistance.

Conclusion

Based on the diverse resistance pattern exhibited by the Staphylococcal isolates carried by wild birds, it can be suggested that wild birds are potential reservoirs and disseminators of resistant strains of Staphylococci and their respective resistance genes into different ecological settings. SCCmec types IV and V of CA-MRS have been studied and reported in wild and domestic animals and are implicated in wide spectrum of community associated infections and increased mortality in both animals and humans. In the current study, high prevalence of CA-MRS SCCmec types VI, V and IV was found. The occurrence of such resistant and potentially infectious strains of Staphylococci in wild birds that are among the elements in least contact with humans is a potential public health concern and it indicates the extent to which these superbugs have disseminated in the environment. This concerns strict surveillance of the reservoirs and drivers of resistance in order to improvise the strategies to control the spread of MRSA in the community.

Statement of funding

This work was partially supported by International Development Research Centre (IDRC; grant 109304-002 under the JPI AMR framework) to RZ and a Ph.D. scholarship to ART by Higher Education Commission (HEC), Pakistan under the project "Aaghaz-e-Huqooq Balochistan initiative indigenous scholarship".

Disclaimer

This study was conducted on the fecal droppings of the wild birds and did not include direct interaction with birds. Hence, no birds/animals were harmed or injured during the study.

CRediT authorship contribution statement

Experimental work, data analysis, and writing of original draft were done by ART while conceptualization, resources, supervision and reviewing and approval of final draft was done by RZ.

Data Availability

All data of the study has been provided in the paper as main figures and in the [supplementary data](#) file.

Declaration of Competing Interest

We hereby declare that we have no conflict of competing interests.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jiph.2023.04.002](https://doi.org/10.1016/j.jiph.2023.04.002).

References

- [1] Aklilu E, Zakaria Z, Hassan, L CH. Molecular relatedness of methicillin-resistant *S. aureus* Isolates from staff. Environ Pets Univ Vet Hosp Malays 2012. PLOS one10.1371/journal.pone.0043329.
- [2] Stryjewski ME, Corey GR. Methicillin-resistant *Staphylococcus aureus*: an evolving pathogen. Clin Infect Dis 2014;58(S1):S10–9. <https://doi.org/10.1093/cid/cit613>
- [3] Lowy FD. *Staphylococcus aureus* infections. New Engl J Med 1998;339:520–32. <https://doi.org/10.1056/nejm199808203390806>

- [4] Turner NA, Batu K, Sharma- Kuinkel S, Maskarinec SA, Eichenberger EM, Shah PP, et al. Methicillin- resistant *Staphylococcus aureus*: an overview of basic and clinical research. *Nat Rev* 2019;41579–018-0147-4. <https://doi.org/10.1038/s41579-018-0147-4>
- [5] Gajdác M. The continuing threat of methicillin-resistant *Staphylococcus aureus*. *Antibiotics* 2019;8:52. <https://doi.org/10.3390/antibiotics8020052>
- [6] Rayner C, Munckhof WJ. Antibiotics currently used in the treatment of infections caused by *Staphylococcus aureus*. *Intern Med J* 2005;35:S3–16. <https://doi.org/10.1111/j.1444-0903.2005.00976.x>
- [7] Grema HA, Geidam YA, Gadzama GB, Ameh JA, Suleman A, et al. Methicillin resistant *Staphylococcus aureus* (MRSA): a review. *Adv Anim Vet Sci* 2015;3(2):79–98. <https://doi.org/10.14737/journal.aavs/2015/3.2.79.98>
- [8] Heaton CJ, Gerbig GR, Sensius LD, Patel V, Smith T. *Staphylococcus aureus* epidemiology in wildlife: a systematic review. *Antibiotics* 2020;9:89. <https://doi.org/10.3390/antibiotics9020089>
- [9] Kale P, Dhawan B. B. The changing face of community-acquired methicillin-resistant *Staphylococcus aureus*. *Indian J Med Microbiology* 2016;3:275–85. <https://doi.org/10.4103/0255-0857.188313>
- [10] Meza MEV, López MG, Quiróz BC, Aranda CMA. Antimicrobial resistance: one Health approach. *Vet World Mar* 2022;15(3):743–9. <https://doi.org/10.14202/2Fvetworld.2022.743-749>
- [11] Sundin GW, Wang N. Antibiotic resistance in plant-pathogenic bacteria. *Annu Rev Phytopathol* 2019;56:161–80. <https://doi.org/10.1146/annurev-phyto-080417-045946>
- [12] Arnold KE, Williams NJ, Bennett M. 'Disperse abroad in the land': the role of wildlife in the dissemination of antimicrobial resistance. *Biol Lett* 2016;11(12):e0168433. <https://doi.org/10.1098/rsbl.2016.0137>
- [13] Monecke S, Widen DG, Hotzel H, Peter M, Guenther S, Lazaris A, et al. Diversity of *Staphylococcus aureus* isolates in European wildlife. *PLoS One* 2016;11(12):e0168433. <https://doi.org/10.1371/journal.pone.0168433>
- [14] Hasan B, Olsen B, Aslam A, Akter L, Melhus A. Dissemination of multidrug resistant ESBL-producing *E. coli* O25b-ST131 clone and role of house crow (*Corvus splendens*) foraging on hospital waste in Bangladesh. *Clin Microbiol Infect* 2015;21(1000):e1–4. <https://doi.org/10.1016/j.cmi.2015.06.016>
- [15] Fahrenfeld N, Ma Y, O'Brien M, Pruden A. Reclaimed water as a reservoir of antibiotic resistance genes: distribution system and irrigation implications. *Front Microbiol* 2013;4:130. <https://doi.org/10.3389/fmicb.2013.00130>
- [16] Hiramatsu K, Kihara H, Yokota H. Analysis of borderline-resistant strains of methicillin-resistant *Staphylococcus aureus* using polymerase chain reaction. *Microbiol Immunol* 1992;36(5):445–53. <https://doi.org/10.1111/j.1348-0421.1992.tb02043.x>
- [17] Al-Talib H, Yean CY, Al-Khateeb A, Hassan H, Singh K-KB, Al-Jashamy K, et al. A pentaplex PCR assay for the rapid detection of methicillin-resistant *Staphylococcus aureus* and Pantone-Valentine Leucocidin. *BMC Microbiol* 2009;9:113. <https://doi.org/10.1186/1471-2180-9-113>
- [18] Oliveira DC, Lencastre H. Multiplex PCR strategy for rapid identification of structural types and variants of the *mec* element in methicillin-resistant *Staphylococcus aureus*. *Antimicrob Agents Chemother* 2002;46(7):2155–61. <https://doi.org/10.1128/aac.46.7.2155-2161.2002>
- [19] Milheiro C, Oliveira DC, de Lencastre H. Multiplex PCR strategy for subtyping the staphylococcal cassette chromosome *mec* type IV in methicillin-resistant *Staphylococcus aureus*: 'SCCmec IV multiplex'. *J Antimicrob Chemother* 2007;60(1):42–8. <https://doi.org/10.1093/jac/dkm112>
- [20] Timsina R, Shrestha U, Singh A, Timsina B. Inducible clindamycin resistance and *erm* genes in *Staphylococcus aureus* in school children in Kathmandu, Nepal. *Future Sci* 2020. <https://doi.org/10.2144/fsoa-2020-0092>
- [21] Zmantar T, Kouidhi B, Miladi H, Bakhrouf A. Detection of macrolide and disinfectant resistance genes in clinical *Staphylococcus aureus* and coagulase-negative *Staphylococci*. *BMC Res Notes* 2011;4:453. <https://doi.org/10.1186/1756-0500-4-453>
- [22] Stepanovic S, Vukovic D, Hola V, Bonoventura GD, Djukic S, Cirkovic I, et al. Quantification of biofilm in microtiter plates: overview of testing conditions and practical recommendations for assessment of biofilm production by *Staphylococci*. *APMIS* Aug 2007;115(8):891–9. https://doi.org/10.1111/j.1600-0463.2007.apm_630.x
- [24] Zaheer Z, Hussain I, Rahman SU, Younas T. Occurrence and antibiotic susceptibility of methicillin-resistant *Staphylococcus aureus* recovered from oropharynx of live cockerels. *Pak Vet J* 2017;37(1):108.
- [25] Zaman S, Sabur A, Hussain J, Hussain P, Amrita. Molecular detection of methicillin-resistant *Staphylococcus aureus* (MRSA) in ornamental birds having public health significance. *J Bangladesh Agric Univ* 2020;18:415–20.
- [26] Lina V, Marius V., Rita S. Prevalence of multi-resistant *Staphylococcus* spp. in faeces of urban birds. 2016; European Congress of Clinical Microbiology and Infectious Diseases; Managing infections promoting science". The Netherlands, Amsterdam, 9–12 Organised by Kenes International. Amsterdam: Kenes International.
- [27] Thapaliya D, Dalman M, Kadariya A, Little K, Mansell V, Taha M. Characterization of *Staphylococcus aureus* in goose feces from state Parks in Northeast Ohio. *EcoHealth* 2017;14(2):303–9. <https://doi.org/10.1007/s10393-017-1227-z>
- [30] Konicek C, Vodrazka P, Bartak P, Knotek Z, Hess C, Racka K, et al. Detection of zoonotic pathogens in wild birds in the cross-border region Austria-Czech Republic. *J Wildl Dis* 2016;52(4):850–61. <https://doi.org/10.7589/2016-02-038>
- [31] Gómez P, Lozano C, Camacho M, Barbero JF, Hernandez JM, Zarazaga M, et al. Detection of MRSA ST3061-t843-mecC and ST398-t011-mecA in white stork nestlings exposed to human residues. *J Antimicrob Chemother* 2016;71(11):53–7. <https://doi.org/10.1093/jac/dkv314>
- [32] Ripa LR, Gomez P, Alonso CA, Camacho MC, Puente JD, Fernandez R. Fet al. Detection of MRSA of Lineages CC130-mecC and CC398-mecA and *Staphylococcus delphini*-Inu (A) in Magpies and Cinereous Vultures in Spain. *Microb Ecol* 2019;78(2):409–15. <https://doi.org/10.1007/s00248-019-01328-4>
- [33] Gills S. *aureus* genomic DNA isolation, on open wet ware, <http://www.openwetware.org/wiki/Gill:S>, accessed May(C), {C}15{C}, {C}2017.
- [34] Lade H, Park JH, Chung SH, Kim IH, Kim JM, Soo JH, et al. Biofilm formation by *staphylococcus aureus* clinical isolates is differentially affected by glucose and sodium chloride supplemented culture media. *J Clin Med* 2019;8(11):1853. <https://doi.org/10.3390/jcm8111853>
- [35] Al-Talib H, Yean CY, Al-Khateeb A, Hassan H, Singh K-KB, Al-Jashamy K, et al. A pentaplex PCR assay for the rapid detection of methicillin-resistant *Staphylococcus aureus* and Pantone-Valentine Leucocidin. *BMC Microbiol* 2009;9:113. <https://doi.org/10.1186/1471-2180-9-113>
- [38] Syed M.A., Jackson C.R., Ramadan H., Afridi R., Shehr Bano S., Bibi S. et al. Detection and Molecular Characterization of *Staphylococcus* from Eggs of Household Chickens. 2019 *FOODBORNE PATHOGENS AND DISEASE*, Volume XX, Number XX. <https://doi.org/10.1089/fpd.2018.2585>
- [39] Bishr AS, Abdelaziz SM, Yahia IS, Yassien MA, Hassouna AN, Aboshanab KM. Association of Macrolide resistance genotypes and synergistic antibiotic combinations for combating Macrolide-resistant MRSA recovered from hospitalized patients. *MDPI* 2021; volume 10(7):10. <https://doi.org/10.3390/biology10070624>
- [40] Rocchetti TT, Martins KB, da Cunha MDL. Detection of the *mecA* gene and identification of *Staphylococcus* directly from blood culture bottles by multiplex polymerase chain reaction. *Braz J Infect Dis* 2018;22(2):99–105. <https://doi.org/10.1016/j.bjid.2018.02.006>
- [41] Magiorakos AP, Srinivasan A, Carey A, Carmeli RB, Falagas Y, C G ME, et al. Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. *Clin Microbiol Infect* 2012;18(3):268–81. <https://doi.org/10.1111/j.1469-0691.2011.03570.x>
- [42] Benrabia I, Hamdi TM, Shehata AA, Neubauer H, Wareth G. Methicillin-Resistant *Staphylococcus aureus* (MRSA) in Poultry Species in Algeria: Long-Term Study on Prevalence and Antimicrobial Resistance. *Vet Sci* 2020;7(2):54. <https://doi.org/10.3390/vetsci7020054>
- [43] Syed MA, Ullah H, Tabbasum S, Fatima B, Woodley TA, Ramadan H, et al. *Staphylococci* in poultry intestines: a comparison between farmed and household chickens, 2020. *Poult Sci* 2020;99(9):4549–57. <https://doi.org/10.1016/2Fj.psj.2020.05.05>
- [44] Jenkins R, Maddocks S. *Bacteriology Methods for the Study of Infectious Diseases*. Academic Press; 2019. <https://doi.org/10.1016/B978-0-12-815222-5.00004-3>